AI in Healthcare: A Glossary of Key Terms

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AI in Healthcare (ITAI 4375) Spring 2026

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March 5, 2026

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Introduction

Healthcare teams are increasingly required to evaluate, implement, and monitor AI-enabled tools across clinical, operational, research, and public health settings. This work necessitates a shared vocabulary among clinicians, data scientists, informaticians, quality and safety leaders, compliance officers, and organizational leadership to ensure that assumptions and risks are discussed with precision. For instance, contemporary "generalist" medical AI approaches prioritize reuse and adaptation across tasks, which can accelerate innovation but also complicate evaluation and oversight if teams lack consensus on valid evidence, safe deployment, and appropriate ongoing monitoring (Moor et al., 2023).

Terminology is critical because many failures in applied healthcare AI are sociotechnical rather than purely technical. Issues such as mismatched data distributions after deployment, ambiguous labeling decisions (reference standards), incomplete reporting, or inadequate lifecycle governance can compromise performance and safety, even when a model demonstrates strong results in development testing (Subasri et al., 2025; Collins et al., 2024). This glossary defines 30 widely used terms from peer-reviewed healthcare AI research, implementation frameworks, and regulatory discussions, and explains how each concept is applied in real healthcare contexts.

Glossary

Term: Active learning

Definition: Active learning is a data-labeling strategy in which a model (or a selection rule) identifies which unlabeled cases would be most informative to label next, so that performance can improve with fewer expert-labeled examples than labeling at random. (Wang et al., 2024).

Example in healthcare: In medical image segmentation, an active-learning pipeline can repeatedly select scans where the current model is most uncertain and request targeted expert annotation, reducing annotation workload while approaching "full-data" performance. (Kim et al., 2024).

Significance in healthcare: Active learning directly addresses one of the biggest constraints in clinical AI development: high-cost expert labeling. When radiologists, pathologists, or nurses must label data, active learning can reduce time and cost, enabling more feasible local development and faster iteration—especially for settings where labeled data are scarce. (Wang et al., 2024).

Term: Algorithmic audit

Definition: An algorithmic audit is a structured review of an AI system's design, data, performance, and governance controls to identify risks and verify that the system meets defined requirements (e.g., safety, reliability, privacy protection, accountability, and documented processes). (National Institute of Standards and Technology [NIST], 2024).

Example in healthcare: Before deploying a hospital AI tool, a governance team may audit documentation on intended use, data provenance, model evaluation, and monitoring plans—aligning the review with an organizational governance framework for healthcare AI. (Hussein et

al., 2026).

Significance in healthcare: Audits translate abstract "trust" requirements into concrete checks. In healthcare, this supports patient safety, regulatory readiness, and organizational accountability by ensuring that tools are not only accurate in a test set, but also governed responsibly across their lifecycle. (NIST, 2024; Hussein et al., 2026).

Term: Annotation

Definition: Annotation is the process of assigning labels or structured markings to data (e.g., images, waveforms, or clinical text) to create "reference" labels for training and evaluation; annotation methods and annotator expertise should be explicitly described. (Tejani et al., 2024).

Example in healthcare: In imaging research, experts may annotate regions of interest (such as lesion boundaries) and the study should report who annotated, how disagreements were handled, and what tools were used. (Tejani et al., 2024).

Significance in healthcare: Annotation quality often determines model ceiling performance: inconsistent or poorly defined labels can produce misleading "accuracy," weaken reproducibility, and obscure whether errors come from the model or the labeling process. Transparent annotation reporting is therefore part of clinical-grade evidence generation. (Tejani et al., 2024).

Term: Area under the receiver operating characteristic curve (AUROC)

Definition: AUROC is a summary measure of discrimination for binary classification; conceptually, it captures how well a model separates cases with and without an outcome across all possible thresholds. (Hospital Pediatrics, 2024).

Example in healthcare: Studies often report AUROC when evaluating a diagnostic classifier or risk model and may show how AUROC changes across validation cohorts or over time. (van

Leeuwen et al., 2025).

Significance in healthcare: AUROC is widely reported because it provides a threshold-independent discrimination metric, but it can be insufficient alone for clinical decision-making—especially in imbalanced datasets—so it is best interpreted alongside calibration and clinical-utility analyses. (Luu et al., 2024; van Leeuwen et al., 2025).

Term: Calibration

Definition: Calibration describes agreement between predicted risks (probabilities) and observed outcome frequencies; a model can have good discrimination yet poor calibration if its predicted probabilities are systematically too high or too low. (Efthimiou et al., 2024).

Example in healthcare: When a risk calculator is moved to a new hospital, it may need recalibration if observed event rates differ from the development setting; calibration plots and calibration-in-the-large are common tools for assessment. (Efthimiou et al., 2024).

Significance in healthcare: Many clinical decisions depend on absolute risk (e.g., treat above a risk threshold). Poor calibration can lead to over-treatment or under-treatment even if AUROC looks strong; therefore, calibration is central to safe deployment and transport of prediction models. (Efthimiou et al., 2024; Ladin et al., 2024).

Term: Change management

Definition: Change management is the systematic approach to preparing people, processes, and workflows for adopting a new tool, including training, stakeholder alignment, usability/workflow redesign, and feedback loops after go-live. (Boag et al., 2024).

Example in healthcare: The “algorithm journey map” approach emphasizes that implementing a healthcare AI tool requires coordinated social and technical work across procurement,

integration, clinical operations, and lifecycle maintenance. (Boag et al., 2024).

Significance in healthcare: Many healthcare AI failures are implementation failures rather than modeling failures. Strong change management improves adoption, reduces workarounds, clarifies accountability, and supports safer real-world use by ensuring tools fit clinical reality. (Boag et al., 2024; van der Vegt et al., 2023).

Term: Clinical data warehouse

Definition: A clinical data warehouse is a centralized repository that integrates heterogeneous healthcare data sources to support reporting, analytics, research, and secondary data use. (Wang et al., 2024).

Example in healthcare: A hospital may use a clinical data warehouse to unify laboratory, imaging, pharmacy, and administrative data for quality reporting and model development/validation workflows. (Wang et al., 2024).

Significance in healthcare: Clinical data warehouses make it feasible to assemble cohorts, evaluate outcomes, and refresh datasets for monitoring. They can materially improve data consistency and accessibility, which supports higher-quality evidence and safer model operations. (Wang et al., 2024).

Term: Computable phenotype

Definition: A computable phenotype is an explicit, operationalized algorithm for identifying patients with a condition or characteristic using electronic healthcare data elements and logical criteria. (Carrell et al., 2024).

Example in healthcare: A research team can define a computable phenotype for a condition by specifying required diagnosis codes, medication patterns, lab thresholds, and temporal logic,

enabling consistent cohort identification across datasets. (Carrell et al., 2024).

Significance in healthcare: Computable phenotypes improve reproducibility in real-world studies and model development because they standardize who is "in the cohort" and how outcomes/exposures are defined—reducing ambiguity and supporting scalable evidence generation. (Carrell et al., 2024).

Term: Confounding

Definition: Confounding occurs when an observed association is distorted because the exposure and the outcome share a common cause; in applied health analytics this can mislead causal interpretation. (Richardson, 2024).

Example in healthcare: In post-deployment evaluation, model-driven clinical interventions can change outcomes, making it difficult to attribute changes in event rates to "model decay" versus improved care—an issue highlighted in postmarket discussions of clinical prediction monitoring. (Ansari et al., 2025).

Significance in healthcare: Confounding is a core reason "real-world performance" can be difficult to measure accurately after deployment. Teams need study designs and monitoring methods that account for changing care processes; otherwise, they may update or retire tools based on misleading signals. (Ansari et al., 2025).

Term: Convolutional neural network (CNN)

Definition: A convolutional neural network is a neural-network architecture that uses convolution operations to learn spatial feature hierarchies, making it well-suited for image analysis tasks. (Chen et al., 2025).

Example in healthcare: CNNs are widely used for medical image classification and

segmentation tasks (e.g., classifying radiographs or segmenting structures in cross-sectional imaging), with extensive research on architectures and evaluation practices. (Chen et al., 2025).

Significance in healthcare: CNNs remain a foundational imaging architecture in clinical AI; understanding their strengths and limitations supports better model selection, interpretation of published evidence, and responsible deployment in imaging workflows. (Chen et al., 2025).

Term: Data governance

Definition: Data governance is the set of policies, roles, and processes that determine how health data are collected, managed, accessed, shared, and protected to support trustworthy and effective digital health systems. (World Health Organization [WHO], 2025).

Example in healthcare: National or organizational governance may define data stewardship responsibilities, permissible uses, quality standards, and transparency mechanisms to enable safe data reuse for advanced analytics. (WHO, 2025).

Significance in healthcare: AI systems are only as trustworthy as the data pipelines behind them. Strong governance supports data quality, privacy protection, and public trust—conditions that are increasingly necessary for sustainable AI adoption across health systems. (WHO, 2025).

Term: Data drift

Definition: Data drift (often discussed within "dataset shift") refers to changes over time in input data distributions or data-generating processes that can degrade model performance after deployment. (Subasri et al., 2025).

Example in healthcare: A monitoring pipeline can detect harmful data shifts in a hospital population and trigger mitigation strategies (e.g., targeted updates) to maintain performance under changing real-world conditions. (Subasri et al., 2025).

Significance in healthcare: Healthcare environments change continuously (coding practices, measurement devices, clinical pathways). Drift-aware monitoring is therefore a safety requirement, not a "nice to have," for many higher-risk AI tools. (U.S. Food and Drug Administration, 2025; Subasri et al., 2025).

Term: De-identification

Definition: De-identification is a privacy-protection process in which health information is transformed so it does not identify an individual and there is no reasonable basis to believe it can be used to identify an individual, consistent with HIPAA standards. (U.S. Department of Health and Human Services, 2025; Electronic Code of Federal Regulations, n.d.).

Example in healthcare: A health system may de-identify datasets before sharing them for multi-institutional research collaborations or analytics development, using HIPAA-permitted methods described in federal guidance. (U.S. Department of Health and Human Services, 2025).

Significance in healthcare: De-identification can enable broader data reuse while reducing privacy risk, which is central to scaling research and innovation. However, healthcare teams must understand de-identification limits and governance requirements to avoid inappropriate data disclosure. (U.S. Department of Health and Human Services, 2025).

Term: Differential privacy

Definition: Differential privacy is a formal approach to privacy that limits what an analysis or model output can reveal about any single individual, typically by introducing carefully calibrated randomness/noise. (Liu et al., 2023).

Example in healthcare: Recent medical-AI work surveys differential privacy methods for training neural network models on sensitive clinical data while preserving useful performance.

(Mohammadi et al., 2026).

Significance in healthcare: Differential privacy offers measurable privacy guarantees and is increasingly relevant for data sharing, model training, and releasing derived datasets. Its practical impact is shaping how teams balance privacy, utility, and equity, especially when models are trained on sensitive patient information. (Liu et al., 2023; Mohammadi et al., 2026).

Term: External validation

Definition: External validation evaluates a prediction model on data that differ from the development data (e.g., different site or time) to assess performance and transportability. (Riley et al., 2024).

Example in healthcare: After developing a model in one institution, a team tests it in another institution or in a later time period and evaluates discrimination, calibration, and clinical usefulness in the new dataset. (Riley et al., 2024).

Significance in healthcare: External validation is a key safeguard against over-optimistic development results and is repeatedly identified as underused but essential for trustworthy clinical models. It supports safer deployment decisions and better understanding of where a model can (and cannot) be used. (Riley et al., 2024; Mansmann, 2025).

Term: Federated learning

Definition: Federated learning is a distributed training approach in which model updates are computed locally at participating sites and aggregated centrally (or collaboratively) without sharing raw patient-level data. (Teo et al., 2024).

Example in healthcare: A systematic review of federated learning in healthcare found that, while the literature is large, only a small fraction of studies reported real-life clinical

application—highlighting the gap between methodological promise and deployment practice. (Teo et al., 2024).

Significance in healthcare: Federated learning can expand access to diverse datasets while improving privacy protections, which is valuable when data sharing is restricted. However, governance, heterogeneity across sites, and implementation complexity can impede real-world clinical uptake. (Teo et al., 2024).

Term: Foundation model

Definition: A foundation model is a large, broadly trained model that can be adapted to many downstream tasks, often using relatively small task-specific datasets compared with training from scratch. (Moor et al., 2023).

Example in healthcare: The "generalist medical AI" paradigm proposes reusable models that can perform diverse medical tasks with limited task-specific labeling, shifting development from building many single-purpose models toward adapting shared pretrained systems. (Moor et al., 2023).

Significance in healthcare: Foundation models can accelerate development and improve scalability across tasks and institutions, but they also raise new oversight challenges—especially around evaluation breadth, transparency of training data, and deployment governance. (Moor et al., 2023; Ranisch & Ranisch, 2025).

Term: Generalizability

Definition: Generalizability is the degree to which a model maintains acceptable performance when used in settings, time periods, or patient populations different from where it was developed. (Subasri et al., 2025).

Example in healthcare: A model can perform well during development but experience performance loss after deployment due to harmful data shifts; monitoring approaches can be used to detect those shifts and protect generalizability in practice. (Subasri et al., 2025).

Significance in healthcare: Generalizability is central to patient safety and equity: if an algorithm works only in one institution or subgroup, it can create uneven care quality. Reporting standards and post-deployment monitoring exist partly to clarify and protect generalizability claims. (Collins et al., 2024; Subasri et al., 2025).

Term: Ground truth

Definition: Ground truth is the "true" reference answer used for training or evaluating a model, typically derived from expert labeling or a defined reference standard; studies should specify how it was established and by whom. (Tejani et al., 2024).

Example in healthcare: In diagnostic modeling, ground truth can come from expert annotations, pathology confirmation, or longitudinal outcomes—each with different limitations that should be transparently described. (Tejani et al., 2024; Drukker et al., 2026).

Significance in healthcare: Ground truth is not automatically "truth"—it can reflect imperfect tests, interrater variability, or biased labeling practices. Because ground truth anchors reported performance, unclear or weak reference standards can overstate effectiveness and mislead clinical teams. (Tejani et al., 2024).

Term: Health information exchange (HIE)

Definition: Health information exchange is the electronic sharing of patient health information among authorized clinicians, organizations, and public health entities to support care coordination and appropriate access to records. (Office of the National Coordinator for Health IT,

2025).

Example in healthcare: HIE can support "push" exchange during transitions of care or "pull" exchange when a receiving system queries for available records, shaping what information is available to clinicians and analytics systems. (Holmgren et al., 2023).

Significance in healthcare: HIE affects data completeness, which affects both care quality and AI performance. More complete, timely patient information can reduce duplication and improve clinical decisions, and it can also improve the fidelity of analytics that rely on longitudinal patient histories. (Office of the National Coordinator for Health IT, 2025; Holmgren et al., 2023).

Term: Human-in-the-loop

Definition: Human-in-the-loop refers to system designs where humans meaningfully participate in steps such as labeling data, reviewing outputs, or making final decisions, rather than fully automating actions. (Salloch & Eriksen, 2024).

Example in healthcare: Ethical analyses argue that healthcare AI should preserve human responsibility and decision authority, clarifying when clinicians (and potentially patients) are the relevant "humans" in oversight loops. (Zheng, 2024).

Significance in healthcare: Human-in-the-loop designs help manage accountability, safety, and trust. They can reduce automation surprises and ensure that clinical judgment remains central, especially when model outputs are uncertain or when consequences are high. (Salloch & Eriksen, 2024; Zheng, 2024).

Term: Institutional review board (IRB)

Definition: An institutional review board is an institutional entity that provides ethical and regulatory oversight of research involving human participants (and, in many contexts, use of

identifiable human data). (National Institutes of Health, 2025).

Example in healthcare: An AI model development study using patient data may require IRB review to assess risks, privacy protections, consent/exemption status, and participant safeguards. (U.S. Department of Health and Human Services, 2021).

Significance in healthcare: IRB oversight is a core mechanism for protecting participants and supporting ethical research conduct. For healthcare AI, IRBs help teams address data privacy, downstream use, and risk/benefit considerations in a structured, regulated way. (National Institutes of Health, 2025).

Term: K-fold cross-validation

Definition: K-fold cross-validation is an internal validation method where data are split into *k* subsets ("folds"); the model is trained on *k–1* folds and evaluated on the remaining fold, repeating across folds to estimate performance. (Collins et al., 2024).

Example in healthcare: Clinical prediction model guidance describes using k-fold cross-validation as a better alternative to a single train/test split for internal performance estimation and model selection. (Efthimiou et al., 2024).

Significance in healthcare: Cross-validation can reduce instability from one-off splits and helps detect overfitting during development. However, it does not replace external validation, which is needed to assess transportability across institutions and time. (Riley et al., 2024; Mansmann, 2025).

Term: Model monitoring

Definition: Model monitoring is the ongoing measurement of an AI system's behavior and performance after deployment, including detection of performance changes over time and

assessment of real-world reliability. (U.S. Food and Drug Administration, 2025).

Example in healthcare: Monitoring pipelines can be designed to detect harmful data shifts and to support mitigation (e.g., targeted updates) while minimizing reliance on expensive manual labels. (Subasri et al., 2025).

Significance in healthcare: Monitoring is increasingly recognized as necessary because clinical environments evolve and feedback loops alter outcomes. Without monitoring, organizations may not notice degradation until harms occur; with monitoring, they can manage lifecycle risk more responsibly. (U.S. Food and Drug Administration, 2025; Ansari et al., 2025).

Term: Multimodal model

Definition: A multimodal model is a model that integrates two or more data modalities (e.g., images, clinical narratives, lab values, physiologic signals) to learn representations and produce outputs informed by multiple sources. (Nature, 2024).

Example in healthcare: A multimodal whole-slide pathology foundation model can combine information from digitized slides and associated medical reports to generate general-purpose representations usable across multiple clinical tasks. (Ding et al., 2025).

Significance in healthcare: Clinical decisions are inherently multimodal; models that better mirror this can improve performance and usability. However, multimodality increases complexity (data alignment, missingness patterns, governance), which raises the importance of thorough evaluation and careful implementation. (Nature, 2024; Ding et al., 2025).

Term: Overfitting

Definition: Overfitting occurs when a model learns patterns specific to the development dataset (including noise) and therefore performs worse when applied to new patients or settings.

(Mansmann, 2025).

Example in healthcare: A model can show strong apparent performance internally, but then fail in external validation datasets; discussions of validation distinguish internal reproducibility from external transportability. (Mansmann, 2025; Riley et al., 2024).

Significance in healthcare: Overfitting can lead to unsafe clinical deployment by overstating effectiveness. Recognizing overfitting drives better methodological choices (robust internal validation, external validation, and realistic performance reporting) and reduces wasted effort rebuilding models that do not transport. (Riley et al., 2024).

Term: Postmarket surveillance

Definition: Postmarket surveillance is the systematic collection and evaluation of evidence after deployment to assess ongoing safety and effectiveness, including detection of performance changes and emerging risks in real-world use. (U.S. Food and Drug Administration, 2024).

Example in healthcare: Policy discussions emphasize that post-deployment assessment can be complicated by real-world clinical interventions that change outcomes, which can mislead monitoring signals if not addressed methodologically. (Ansari et al., 2025).

Significance in healthcare: Postmarket surveillance is a cornerstone of lifecycle safety for higher-risk AI tools. It supports timely detection of harm, informs model updates (or retirement), and aligns real-world evidence collection with healthcare's safety culture and regulatory expectations. (U.S. Food and Drug Administration, 2024; Ansari et al., 2025).

Term: Real-world evidence (RWE)

Definition: Real-world evidence is clinical evidence regarding the usage and potential benefits or risks of a medical product derived from analysis of real-world data sources. (U.S. Food and

Drug Administration, 2026).

Example in healthcare: Regulators publish guidance describing how they evaluate whether real-world data are fit-for-use to generate RWE for regulatory decisions, illustrating how RWE is formalized for oversight contexts. (U.S. Food and Drug Administration, 2025).

Significance in healthcare: RWE is increasingly important for understanding performance, safety, and impact outside controlled studies. For healthcare AI, RWE concepts also shape how organizations think about deployment evidence, postmarket monitoring, and ongoing evaluation in routine practice. (U.S. Food and Drug Administration, 2026).

Term: Synthetic data

Definition: Synthetic data are artificially generated data that aim to mimic statistical properties of real health data while reducing disclosure risks compared with sharing original records. (Giuffrè et al., 2023).

Example in healthcare: Reviews of synthetic medical data emphasize that evaluation must consider both utility and privacy risk; many studies historically under-evaluated privacy risk, motivating calls for more standardized assessment practices. (Kaabachi et al., 2025).

Significance in healthcare: Synthetic data can expand access for development, testing, and education while reducing barriers to data sharing. However, synthetic data are not "automatically safe," and credible evaluation frameworks are needed to prevent underestimated re-identification or leakage risks and to ensure the synthetic data remain clinically useful. (Giuffrè et al., 2023; Kaabachi et al., 2025).

Term: Transformer

Definition: A transformer is a neural-network architecture built around attention mechanisms (self-attention) to model relationships within sequences; it was introduced as an alternative to recurrence and convolution for sequence tasks. (Vaswani et al., 2017).

Example in healthcare: Systematic reviews describe "vision transformer" and hybrid transformer architectures being used for radiology tasks such as classification and segmentation, demonstrating how transformer-style attention mechanisms have expanded beyond language to medical imaging. (Kim et al., 2025).

Significance in healthcare: Transformer architectures underpin many modern foundation-model approaches and are increasingly applied across modalities (text, imaging, multimodal). Understanding transformers helps multidisciplinary teams interpret why certain models scale well, what kinds of data they consume, and what evaluation/monitoring questions arise when such models are adapted for clinical use. (Vaswani et al., 2017; Ding et al., 2025).

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